Supplementary Material: BaitFL

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This document retains the detailed plots, operation-level tables and expanded procedures accompanying paper version 7.2. The main study contains 96 completed runs; the prospective review cases and the single-cycle norm-matching diagnostic are separate evaluations.

S1. Complete Representative Score Geometry

Figures S1 and S2 retain the representative round-20 grid from study seed 42. Coordinates are asinh of mean A/B norm-z and profile-z scores. Each panel contains 100 clients, with coincident points possible. Blue circles denote honest clients and orange triangles free riders. These panels show norm separation; they do not establish its universality across seeds, cycles or magnitude-matched fabrication. The main paper's honest-error figure presents a different seed selected by its observed errors.

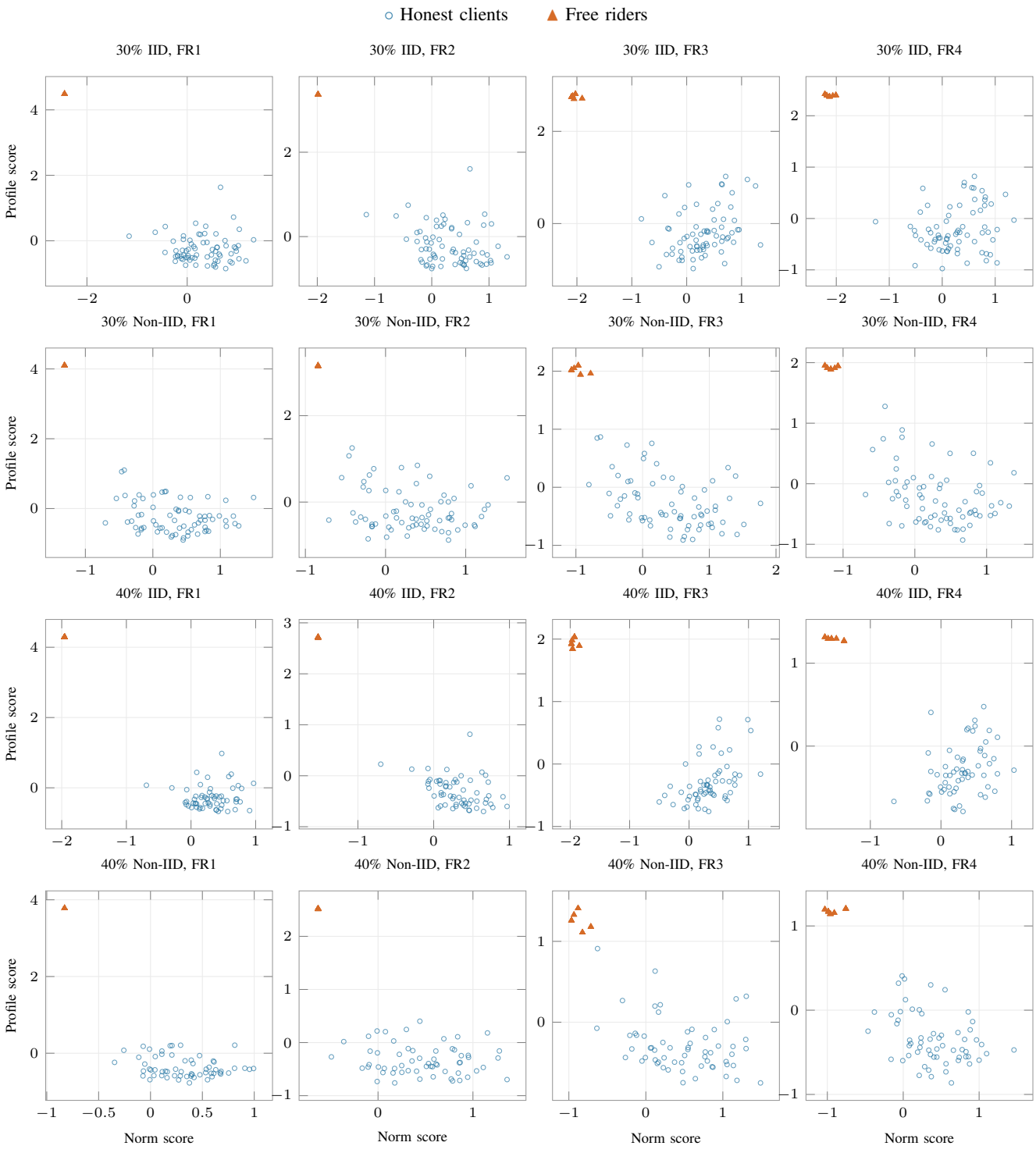


Fig. S1. Representative round-20 MNIST responses. Axes show asinh of paired mean norm and profile z-scores.

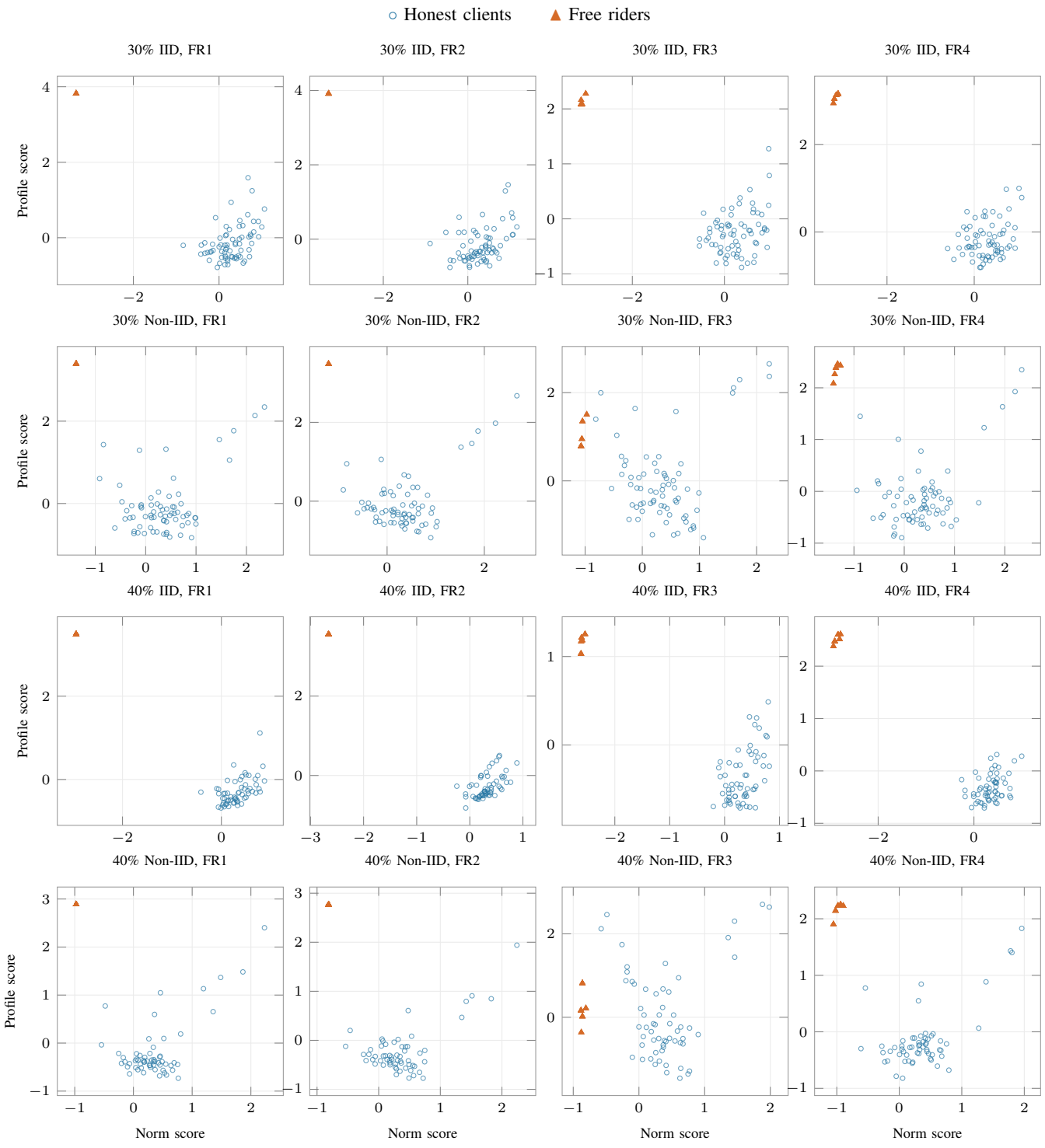


Fig. S2. Representative round-20 CIFAR-10 responses. Axes show asinh of paired mean norm and profile z-scores.

S2. Detailed Comparison and Honest-Client Costs

Table S1 compares published information and computation requirements qualitatively. It is not an empirical ranking. Table S2 retains all attack-level honest-client summaries. Fractional client counts are means across completed runs; ranges describe the observed integer counts in individual runs. An unresolved label alone does not withhold an ordinary response or renew review.

TABLE S1
REQUIREMENTS OF FREE-RIDER AUDITING APPROACHES.

Approach	Extra learning	Additional client work	Server work	Information and exposure
STD-DAGMM [1]	Autoencoder and mixture fitting	No separate peer audit	Feature construction, fitting and density scoring	Individual response features are inspected.
FRAD [2]	DAGMM fitting	Contribution-related inputs	Learned detection and reputation updates	Contribution/resource information supplements response evidence.
PASS [3]	No separate anomaly learner	Evaluate other clients' models and report scores	Coordinate parameter audits and participation	Peer models reach auditing clients, with perturbation/pruning protections.
FRIDA canary variants [4]	Client canary training	Train on server-provided examples	Canary loss or gradient tests	Server holds labeled canaries and inspects individual responses.
FRIDA property variants [4]	Attack-dependent inference	No canary training	Infer local-data characteristics	Inferred dataset properties form the detection evidence.
BaitFL	No auxiliary learning	Train one ordinary or challenge model	Norms, profiles, statistical splits and confirmation	Individual responses remain visible. Challenge responses are withheld from aggregation.

Privacy is compared through required information and exposure, not through measured leakage rates. BaitFL does not provide differential privacy or directly support standard secure aggregation.

TABLE S2
HONEST-CLIENT IMPACT.

(a) Final unresolved clients and withheld responses

Dataset	Partition	FR (%)	Unresolved honest clients				Honest responses withheld (%)			
			FR1	FR2	FR3	FR4	FR1	FR2	FR3	FR4
MNIST	IID	30	1	4	6.33	5.33	11	11	11	11
		40	0.67	5.33	4.33	16.33	11	11	11	11.67
	Non-IID	30	5	4.67	41.67	40.67	11	11	12.32	11.74
		40	15	6.33	37	35	13.18	11	12.86	14.57
CIFAR-10	IID	30	2.33	0	2.33	5.67	11	11	11	11
		40	0.33	0.33	4	4.33	11	11	11	11
	Non-IID	30	17	5	42.67	27.33	11.64	11	15.02	12.16
		40	6	3.67	43	14	11	11	17.08	11.67

(b) Honest-client exclusions

Dataset	Partition	FR (%)	Attack	Honest clients [range]		Excluded client-rounds	
				Permanent removals	Ever quarantined	Quarantine	Ordinary responses
MNIST	Non-IID	30	FR3	0	1.33 [0–3]	26.67	1
		40	FR3	0.67 [0–2]	3.33 [1–5]	60	2.67
			FR4	0.33 [0–1]	1 [0–2]	20	0.67
CIFAR-10	Non-IID	30	FR3	0	0.67 [0–1]	13.33	0
			FR4	0	0.67 [0–2]	13.33	0.67
		40	FR3	0	3.67 [1–9]	73.33	2

S3. OPERATION-LEVEL COMPLEXITY AND PAYLOADS

The main paper defines N, P, L, B, K. Table S3 separates incremental detector work from ordinary training and aggregation. The temporary-state bounds are implementation bounds rather than measured process peaks. Table S4 counts packed numerical payloads; Python objects, clients, attack histories and logging add overhead.

TABLE S3
INCREMENTAL DETECTOR COMPUTATION AND WORKING STATE.

Operation	Work	Working state	Implementation
Norms, profiles and fingerprints	$O(NP)$ per round	$O(P)$ scratch and $O(N)$ hashes	Read returned tensors without training an auxiliary model.
Selected challenge summaries	$O(NL)$ per paired cycle	$O(NL)$	Retain two small profiles per client.
Leave-one-out comparisons	$O(N^2(L + \log N))$ per cycle	$O(NL)$ temporary	Form peer medians and distances for each client.
Histogram and bootstrap splits	$O(B(N + K))$ per cycle	$O(BN + BK)$ temporary	Resample scalar scores across three views.
Geometric tests and confirmation	$O(N)$ per cycle	$O(N)$ counters	Apply fixed-component tests and update evidence memory.
Shared challenge construction	$O(P)$ per cycle	$O(P)$	Generate two shared models.

TABLE S4
CALCULATED DETECTOR PAYLOAD SIZES.

Model	Parameters	P	L	Two traps (MiB)	Pair payload (KiB)	Index array (MiB)	Counts (MiB)
MNIST	206922	206922	8	1.579	8.594	0.763	0.076
CIFAR-10	620810	621258	22	4.740	19.531	0.763	0.076

S4. EXPANDED PUBLISHED-ENGINE PROCEDURES

Algorithms S1–S4 expand the four phases of the frozen v15.4 study engine. The main text provides the score and certificate equations and a compact overall procedure. Protocol settings are unchanged. For traceability, Algorithm S4 retains the published raw-finding clock rule; Section S5 distinguishes its corrected prospective implementation.

Algorithm S1 Phase I: ordinary warmup

procedure WARMUP($w, \mathcal{S}$)
  for $t = 1, \ldots, T_{\text{warm}}$ **do**
    Send w to all active clients
    Collect responses, summaries and send-back checks
    $\mathcal{A}_t \leftarrow$ all active clients
    $w \leftarrow$ AGGREGATE($w, \mathcal{A}_t$)
  end for
  return $(w, T_{\text{warm}}, \mathcal{S})$
end procedure

Algorithm S2 Phase II: collect and assess paired responses

procedure PAIREDCYCLE($w, t, \mathcal{S}, c, \sigma,$ initial)
  Create fixed independent traps A, B from a copy of w
  if initial **then**
    Shuffle groups and schedule one response to each trap
  else
    Group active-review and remaining clients separately
    Alternate traps for reviewed clients throughout the cycle
    Schedule each remaining client once on each trap
  end if
  Preselect one position per client and trap
  Initialize selected summaries O
  $h \leftarrow \min(\ell_{\text{cyc}}, T - t)$
  for $s = 1, \ldots, h$ **do**
    Advance t and read trap recipients $\mathcal{T}_t$
    Send the trap to $\mathcal{T}_t$ and w to others
    Collect responses and retain preselected summaries
    $\mathcal{A}_t \leftarrow$ ordinary, non-quarantined recipients at round start
    if $s = \ell_{\text{cyc}}$ **then**
      Require complete A/B coverage of active clients
      $\mathcal{E} \leftarrow$ SCREENANDCERTIFY(O)
      $\mathcal{S} \leftarrow$ CONFIRM($\mathcal{S}, \mathcal{E}, c$)
      Exclude new removals and quarantines from $\mathcal{A}_t$
    end if
    $w \leftarrow$ AGGREGATE($w, \mathcal{A}_t$)
  end for
  if the cycle completed **then**
    Advance c and clear the completed schedule
    $\sigma \leftarrow t$ if no review/probation remains, else $\perp$
  end if
  return $(w, t, \mathcal{S}, c, \sigma)$
end procedure

Algorithm S3 Phase III: confirm evidence and bound review

procedure CONFIRM($\mathcal{S}, \mathcal{E}, c$)
  Require one complete decision per active client and a newer cycle
  for all active clients i **do**
    $J_i \leftarrow$ statistical components certified on both traps
    Advance or reset the consecutive paired send-back streak
    if the cycle result is clear **then**
      Advance the consecutive-clear certificate-clean streak
      Clear $M_{i,n}, M_{i,p}$ after two consecutive clear cycles
    else
      Reset the certificate-clean streak
      Increment $M_{i,j}$ once for each fresh $j \in J_i$
    end if
    if a fresh component reaches two, or send-back is confirmed **then**
      Permanently remove i
    else if strong paired evidence is present **then**
      Quarantine i and renew its review budget to two cycles
      Reset recovery, probation and cooldown
    else
      Decrease the review budget toward zero
      Apply consecutive-clear rehabilitation and probation
      if unconfirmed review has expired **then**
        Release i and set a two-audit cooldown
      end if
    end if
  end for
  return $\mathcal{S}$
end procedure

Algorithm S4 Phase IV: ordinary rounds and sparse audits

procedure SPARSEROUND($w, t, \mathcal{S}, \sigma$)
  $t \leftarrow t + 1$
  if active clients remain, $\sigma \neq \perp$ and $t - \sigma \geq I$ **then**
    Send one fresh trap to all active clients
    Collect same-trap norm/profile scores and send-back checks
    $\mathcal{F} \leftarrow$ clients with a single-trap certificate or send-back
    $\mathcal{D} \leftarrow$ clients currently in cooldown
    $\mathcal{F}_{\text{elig}} \leftarrow \mathcal{F} \setminus \mathcal{D}$
    Decrease all audit cooldowns toward zero
    for all $i \in \mathcal{F}_{\text{elig}}$ **do**
      Open two cycles of review without quarantine
      Reset clean progress and the paired send-back streak
    end for
    $\sigma \leftarrow t$ if $\mathcal{F} = \emptyset$, otherwise $\perp$
    Keep w unchanged
  else
    Send w to active clients and collect responses
    $\mathcal{A}_t \leftarrow$ eligible ordinary clients
    $w \leftarrow \text{AGGREGATE}(w, \mathcal{A}_t)$
  end if
  return $(w, t, \mathcal{S}, \sigma)$
end procedure

S5. AUDIT-CLOCK TRACE CHECK

The frozen engine restarts the quiet clock when the raw finding set $\mathcal{F}$ is empty. Audit enrollment instead uses $\mathcal{F}_{\text{elig}} = \mathcal{F} \setminus \mathcal{D}$, where $\mathcal{D}$ is the pre-decrement cooldown set. If all findings are in cooldown, the raw guard can unset the clock without opening review and thereby prevent later audits. The corrected review engine restarts the clock when no actual review or probation requires paired coverage.

The guards agree on all 622 recorded audits in the 96-run study. There are 609 empty-finding audits. In each of the other 13, the raw finding count exceeds the number of active clients in cooldown, proving that at least one finding remains eligible. This is a trace agreement check, not equivalence for unobserved cooldown states. The primary training results were not relabeled or recomputed. Prospective honest-only and low-prevalence runs use the corrected guard and an independent registered source manifest.

S6. REPRODUCIBILITY AND SCOPE OF THE NEW DIAGNOSTIC

The norm-matching export contains 72 round-20 FR2–FR4 cases. The analysis first reproduces every stored statistical certificate with the unchanged decision backend. It then replaces only attacker norms by the same-trap median honest norm while preserving the normalized tensor profile. Evaluation labels construct this oracle input; the detector receives no labels. This tests feature sensitivity under additional attacker information, not a complete attack execution or confirmed removal. The full per-case counts, completed-run inventory, queue snapshot and input hashes are in `generated/revision/`.

REFERENCES

[1] J. Lin, M. Du, and J. Liu, "Free-riders in federated learning: Attacks and defenses," arXiv preprint arXiv:1911.12560, 2019. [Online]. Available: https://arxiv.org/abs/1911.12560

[2] B. Wang, H. Li, X. Liu, and Y. Guo, "FRAD: Free-rider attacks detection mechanism for federated learning in AIoT," *IEEE Internet of Things Journal*, vol. 11, no. 3, pp. 4377–4388, 2024. [Online]. Available: https://doi.org/10.1109/JIOT.2023.3298606

[3] J. Wang, X. Chang, J. Mišić, V. B. Mišić, and Y. Wang, "PASS: A parameter audit-based secure and fair federated learning scheme against free-rider attack," *IEEE Internet of Things Journal*, vol. 11, no. 1, pp. 1374–1384, 2024. [Online]. Available: https://doi.org/10.1109/JIOT.2023.3288936

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